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"99.9% of science is boring."
- Makise Kurisu

PRECALCULUS

July 28, 2026

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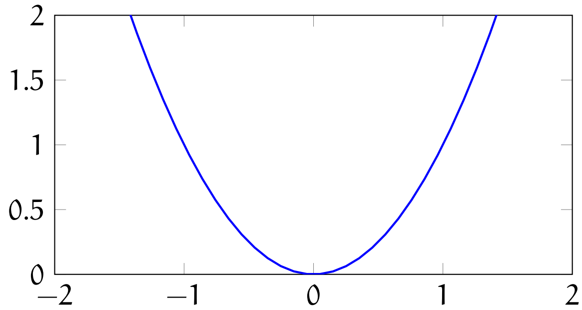
CONTENTS

3.7 Graphs of Quadratics

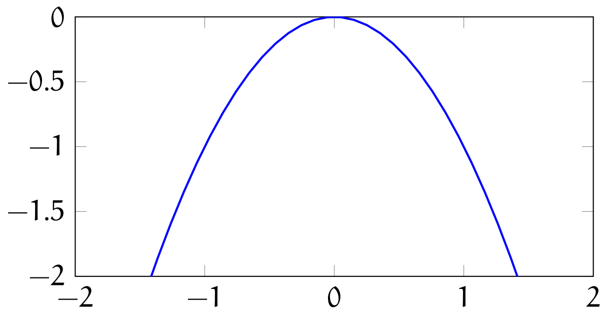
Definition 3.8 (Graphing Quadratics). The process of drawin a U-shaped curve called a *parabola* on a grid based on an equation where the highest power of the variable is 2 ($f(x) = ax^2 + bx + c$).

Note: The graph will always look like if a is.

1. if $a > 0$ it becomes an upward opening parabola.



2. if $a < 0$ it becomes a downward opening parabola.



Definition 3.9 (y Intercept). The point where the graph (a U-shaped curve called a parabola) crosses the vertical y-axis, always occuring where $x = 0$. The y-intercept is simply the constant number c.

$$(0, c)$$

Definition 3.10 (x Intercept). A point where the graph (a U-shaped curve called a parabola) crosses the horizontal x-axis.

$$ax^2 + bx + c = 0$$

Definition 3.11 (Vertex). The turning point on a parabola's U-shaped graph where the function reaches its maximum or minimum value, and the graph changes direction.

$$\left(\frac{b}{2a}, f(x)\right)$$

Or we can complete the square for the vertex form.

Example 3.11 (Quadratics Graphs).

1. $f(x) = 2x^2 + 8x + 5$

(a) Vertex:
$\left(-\frac{(8)}{2(2)}, \quad\right)$
Note: We plug in -2 into the function.
$(-2, -3)$

(b) y-intercept:
$(0, 5)$

(c) x-intercept:
$2x^2 + 8x + 5 = 0$
$x = \frac{-(8) \pm \sqrt{(8)^2 - 4(2)(5)}}{2(2)}$
$x = \frac{-8 \pm \sqrt{24}}{4}$
$x \approx -3.22 \wedge x \approx -0.77$